CT-WM: A Continuous Thought World Model for Long-Distance Sparse Rewards Autonomous UAV Navigation

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Abstract

World Models (WMs) have shown considerable potential in reinforcement learning (RL)-based autonomous unmanned aerial vehicles (UAVs) navigation, where the learned latent dynamics provide predictive imagination for downstream planning. Conventional world models, such as RSSM-based architectures, recursively maintain historical observations, actions, and state uncertainty within a compact latent representation. However, in long-horizon autonomous UAVs navigation under sparse-reward settings, the agent often receives limited task feedback before reaching the goal, colliding with obstacles, or violating safety constraints. As a result, the finite-dimensional latent state must continuously compress extended sensory inputs, control actions, and delayed environmental responses, which may cause early but task-relevant navigation cues to be gradually weakened during recurrent updates. To address this limitation, we propose a continuous-thought-inspired world model that replaces the instantaneous deterministic transition in conventional RSSM with a dynamic synchronization-based latent context. Through internal thought ticks, neuron-level temporal modeling, and neural synchronization representations, the proposed architecture expands historical information from a single compressed deterministic state into an ongoing internal reasoning process. This design improves sparse-reward UAVs navigation by strengthening long-horizon latent belief modeling and physical risk perception, enabling the world model to move beyond one-step prediction toward continuous internal imagination for autonomous UAVs decision-making.

1 Introduction

In recent years, the rapid development of the Internet of Things (IoT) has promoted the deployment of unmanned aerial vehicles (UAVs) in a wide range of intelligent service scenarios, including logistics delivery, emergency rescue, last-mile distribution, agricultural monitoring, precision irrigation, and human–machine collaborative tasks. The realization of these intelligent systems relies not only on perception and control, but also on the ability of robots to navigate autonomously, learn from task feedback, and interact with local environments. Compared with constrained environments, UAVs navigation in open areas has achieved considerable progress, partly because open spaces impose fewer spatial restrictions and make physical risk prediction relatively tractable. However, indoor UAVs navigation remains challenging. Dense obstacles, narrow passages, and complex spatial layouts restrict the maneuverability of UAVs and increase the difficulty of predicting physical risks during flight.

Traditional UAVs navigation methods commonly rely on simultaneous localization and mapping (SLAM), together with perception and obstacle-avoidance modules, to support autonomous navigation. Traditional non-learning-based navigation methods are often effective in relatively simple and small-scale environments, mainly because the surrounding geometric structure can be explicitly modeled and local obstacles can be avoided through predefined rules or optimization strategies. However, such methods usually rely heavily on accurate perception, map construction, and manually designed decision rules. As the environment becomes larger and physically more complex, these assumptions become harder to satisfy. In densely cluttered regions, continuously constructing and updating maps may introduce additional computational and planning costs, which can reduce navigation efficiency. Moreover, handcrafted perception-and-avoidance pipelines are often less adaptable to unseen layouts, environmental changes, and different task conditions.

2 Method

2.1 Modeling and Formula Expression

We formulate CT-WM around a continuous-thought recurrent state-space model, denoted as CT-RSSM. Conventional RSSM updates a deterministic state

as an instantaneous recurrent snapshot, e.g., $\mathbf{h}_t = f_\theta(\mathbf{h}_{t-1}, \mathbf{z}_{t-1}, \mathbf{a}_{t-1})$. This design is compact, but in long-horizon sparse-reward navigation the same finite-dimensional state must repeatedly overwrite observations, actions, and delayed task cues. Early but decisive evidence can therefore be diluted before the downstream reward or safety signal becomes available. CT-RSSM replaces the instantaneous deterministic state $\mathbf{h}_t$ with a synchronization context $\bar{\mathbf{c}}_t$ generated by an internal continuous-thought process. The model strengthens long-horizon credit backtracking along three axes: the representation axis uses neural synchronization rather than a single hidden snapshot; the neuron axis assigns each neuron a private neuron-level temporal model; and the temporal axis unfolds multiple internal thought ticks inside each environment step.

Let t denote the external environment step and let $\tau \in \{1, \dots, K\}$ denote an internal thought tick. The stochastic latent state is $\mathbf{z}_t$, the action is $\mathbf{a}_t$, the observation is $\mathbf{x}_t$, and $\mathbf{c}_t^+$ is the posterior synchronous belief after assimilating $\mathbf{x}_t$. Before observing $\mathbf{x}_t$, CT-RSSM constructs a prior synchronous context $\bar{\mathbf{c}}_t$ from the previous posterior belief, previous stochastic latent, and previous action. The CT-RSSM core is defined as follows.

The sequence model constructs the prior context:

$$\bar{\mathbf{c}}_t = \mathrm{SyncCTM}_\theta(\mathbf{c}_{t-1}^+, \mathbf{z}_{t-1}, \mathbf{a}_{t-1}). \tag{1}$$

$$\boldsymbol{\alpha}_{t,\tau} = f_\theta^{\mathrm{syn}}(\boldsymbol{\nu}_{t,\tau-1}, \mathbf{z}_{t-1}, \mathbf{a}_{t-1}, \mathbf{o}_{t,\tau}). \tag{2}$$

$$\nu_{t,\tau,d} = g_{\theta_d}(\alpha_{t,\tau-M+1:\tau,d}). \tag{3}$$

$$S_{t,\tau}^{ij} = \frac{\sum_{k=1}^{\tau} \exp[-r_{ij}(\tau-k)]\, \nu_{t,k,i}\nu_{t,k,j}}{\sqrt{\sum_{k=1}^{\tau} \exp[-r_{ij}(\tau-k)]}}. \tag{4}$$

Here, $d = 1, \dots, D$. The synchronization state is projected into a context and used by the dynamics and encoder:

$$\bar{\mathbf{c}}_{t,\tau} = \phi_\theta(\{S_{t,\tau}^{ij}\}_{(i,j)\in\mathcal{P}}), \tag{5}$$

$$\bar{\mathbf{c}}_t = \bar{\mathbf{c}}_{t,K}. \tag{6}$$

$$\hat{\mathbf{z}}_t \sim p_\theta(\hat{\mathbf{z}}_t \mid \bar{\mathbf{c}}_t). \tag{7}$$

$$\mathbf{z}_t \sim q_\theta(\mathbf{z}_t \mid \bar{\mathbf{c}}_t, \mathbf{x}_t), \tag{8}$$

$$\mathbf{c}_t^+ = \eta_\theta(\bar{\mathbf{c}}_t, \mathbf{z}_t, e_\theta(\mathbf{x}_t)). \tag{9}$$

Here, $\boldsymbol{\alpha}_{t,\tau}$ is the synaptic pre-activation vector, $\boldsymbol{\nu}_{t,\tau}$ is the post-activation vector, M is the neuron-level memory window, D is the number of neurons, $\mathcal{P}$ is the selected set of neuron pairs, and $r_{ij} \geq 0$ is a learnable decay rate for pair (i, j). The vector $\mathbf{o}_{t,\tau}$ denotes an optional read or attention signal within the thought tick; in pure latent imagination it can be set to a learned null vector. The private NLM g_{θ_d} prevents all dimensions from sharing a single overwrite rule, so a neuron carrying rare but task-critical evidence can preserve a distinct temporal trace across the internal tick sequence.

The synchronization state $S_{t,\tau}^{ij}$ is the key replacement for an instantaneous hidden state. Instead of treating $\mathbf{h}_t$ as the only carrier of history, CT-RSSM represents belief by pairwise temporal correlations among neuron-level activities. The decay factor allows different pairs to specialize to short-, medium-, or long-range internal dependencies. As a result, the latent prior $p_\theta(\hat{\mathbf{z}}_t \mid \bar{\mathbf{c}}_t)$ is conditioned on a synchronized thought trajectory, while the posterior encoder $q_\theta(\mathbf{z}_t \mid \bar{\mathbf{c}}_t, \mathbf{x}_t)$ injects the current observation only after the prior context has been formed.

2.2 Within-Step Credit Re-Estimation

The internal thought ticks expose intermediate contexts $\bar{\mathbf{c}}_{t,\tau}$ before the final context $\bar{\mathbf{c}}_t$ is selected. We use these intermediate contexts to re-estimate credit inside each environment step. For each tick, a latent consistency cost is defined as

$$\ell_{t,\tau}^{\mathrm{lat}} = D_{\mathrm{KL}}[q_\theta(\mathbf{z}_t \mid \bar{\mathbf{c}}_{t,\tau}, \mathbf{x}_t) \,\|\, p_\theta(\mathbf{z}_t \mid \bar{\mathbf{c}}_{t,\tau})]\,. \tag{10}$$

The model then assigns a differentiable credit weight to every tick:

$$\omega_{t,\tau} = \frac{\exp(-\lambda \ell_{t,\tau}^{\mathrm{lat}})}{\sum_{\kappa=1}^{K} \exp(-\lambda \ell_{t,\kappa}^{\mathrm{lat}})}, \tag{11}$$

$$\mathcal{L}_{\text{CT-RSSM}} = \sum_{t=1}^{T} \sum_{\tau=1}^{K} \omega_{t,\tau} \ell_{t,\tau}^{\mathrm{lat}}, \tag{12}$$

where λ controls how sharply credit concentrates on the most explanatory internal ticks. This objective does not force the last tick to absorb all responsibility. Instead, gradients can flow to the synapse model, the private NLMs, and the synchronization pairs that formed useful intermediate beliefs. The resulting credit path can be written as

$$\frac{\partial \mathcal{L}_t}{\partial \theta_d} = \sum_{\tau=1}^{K} \sum_{(i,j)\in\mathcal{P}} \omega_{t,\tau} \frac{\partial \ell_{t,\tau}^{\mathrm{lat}}}{\partial \bar{\mathbf{c}}_{t,\tau}} \frac{\partial \bar{\mathbf{c}}_{t,\tau}}{\partial S_{t,\tau}^{ij}} \frac{\partial S_{t,\tau}^{ij}}{\partial \nu_{t,1:\tau,d}} \frac{\partial \nu_{t,1:\tau,d}}{\partial \theta_d}. \tag{13}$$

Equation (13) makes the credit-backtracking mechanism explicit: a delayed latent error at environment step t is redistributed over the internal tick trajectory and then routed to the neuron-level temporal histories that produced the synchronized representation. Therefore, CT-RSSM lengthens the effective credit path from a single recurrent transition to a structured sequence of K differentiable thought updates per environment step.

2.3 Prediction Heads Outside CT-RSSM

Reward prediction, termination prediction, and observation reconstruction are not part of the CT-RSSM core in Equations (1)–(9). They are external task heads attached to the posterior belief for training and planning supervision:

$$\text{Decoder:} \quad \hat{\mathbf{x}}_t \sim p_\psi^{\mathrm{dec}}(\hat{\mathbf{x}}_t \mid \mathbf{c}_t^+, \mathbf{z}_t). \tag{14}$$

$$\text{Reward predictor:} \quad \hat{r}_t \sim p_\psi^{r}(\hat{r}_t \mid \mathbf{c}_t^+, \mathbf{z}_t). \tag{15}$$

$$\text{Termination predictor:} \quad \hat{d}_t \sim p_\psi^{d}(\hat{d}_t \mid \mathbf{c}_t^+, \mathbf{z}_t). \tag{16}$$

The decoder, reward predictor, and termination predictor provide reconstruction, reward, and done-signal losses, but they do not define the recurrent synchronization transition itself. This separation keeps CT-RSSM

focused on long-horizon latent dynamics and credit-preserving belief construction, while allowing downstream heads to be replaced or ablated without changing the core world-model transition.

3 Experiments

Experimental results, baselines, implementation details, and ablation studies should be added here.

4 Conclusion

This paper presents CT-WM, a continuous-thought-inspired world model for long-distance sparse-reward autonomous UAV navigation. By replacing an instantaneous deterministic recurrent transition with a synchronization-based latent context, CT-RSSM provides a structured internal reasoning trajectory for preserving long-horizon navigation cues and routing delayed credit through neuron-level temporal histories.